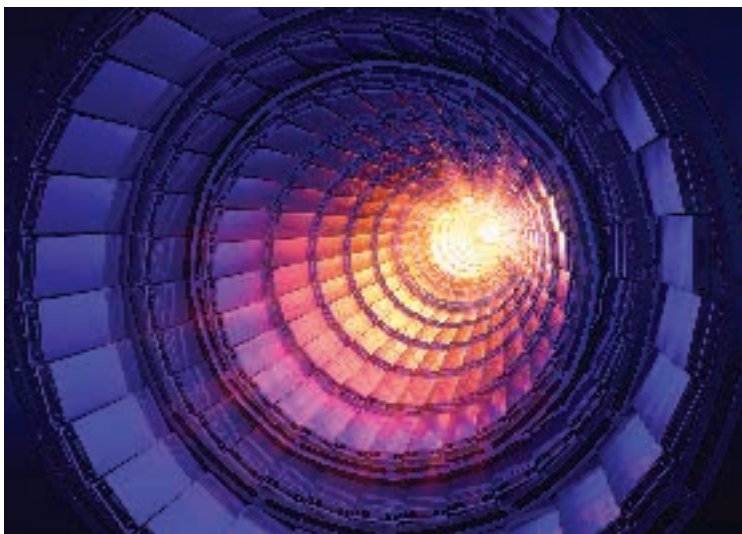




small fraction of them are used in fundamental particle research. Most are used for industrial purposes such as ion implantation (crucial to semiconductor fabrication) and radiation therapy (treating cancer). However, apart from these other uses the most exciting is still their role in unravelling the mysteries of matter. Particle accelerators are fundamental in experimental particle physics which quite literally gives us answers to the very existence and nature of matter, the origins of the big bang and the structure of the subatomic universe.

Penicillin

Ingesting fungi, in order to kill bacteria. That's the basic idea that Penicillin, the world's first antibiotic, works on. Sounds perverse, but the discovery of penicillin has increased the lifespan of humans from 20-30 years to what it is today.

In 1928, Sir Alexander Fleming observed that colonies of the bacterium *Staphylococcus aureus* could be destroyed by the mould *Penicillium notatum*, proving that it contained an antibacterial agent. This observation, later led to medicines that could attack and kill certain types of disease-causing bacteria inside the body. This is why the discovery of penicillin is believed by many to have been the start of chemical warfare on a microscopic scale.